

Terrestrial Air Temperature: 1900-2017 Gridded Monthly Time Series

(Version 5.01)

interpolated and documented by

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STATION DATA SOURCES:

Station data, weather station-monthly-mean air temperature (T , °C), were compiled from several updated sources including a recent version of the Global Historical Climatology Network Monthly (GHCNM) Version 3 ([GHCN3](#)) dataset (Lawrimore et al., 2011); the Daily Global Historical Climatology Network ([GHCN-Daily](#)) archive (Menne et al., 2012); the Atmospheric Environment Service/Environment Canada archive; records of the State Hydrometeorological Institute, St. Petersburg, Russia; data for Greenland--taken from the GC-Net (Steffen et al., 1996); records from the Automatic Weather Station Project (courtesy of Charles R. Stearns at the University of Wisconsin-Madison); the Global Synoptic Climatology Network archive (Dataset 9290c, courtesy of National Climatic Data Center); and observations contained within the Global Surface Summary of Day ([GSOD](#)).

Some of the T archives consisted of daily observations and monthly T values were derived from them. A monthly station value was calculated from the daily station observations, only when the number of missing daily observations for that month was no more than five. If more than five daily observations were missing, the monthly value was encoded as missing. In the past, within the GSOD archive, we observed that some of the daily air-temperature observations appeared to be unrealistic. To mitigate potentially deleterious influences from unrealistic records, we first applied several filters (similar to filters described by Durre et al., 2010) to the original GSOD daily values. This filtering helped remove many of the unrealistic records, including duplicated months and years.

Many of the available station records were merged to create composite station-record series. Station records located within 2.5 km of each other were combined and subsequently treated as an individual station record. In order to reduce the influences of outliers, the median of each set of same-month values was taken as that month's composite station-record value. The location coordinates of each composite station record were taken as the averages of the merged stations' two geographic coordinates. To check for unusual values in each composite time series, the estimated values were compared with monthly climatological norms estimated by Legates and Willmott (1990). Legates and Willmott (LW) station norms were interpolated to each composite station-record location and the absolute difference between each monthly composite-station value and the

estimated monthly LW norm was calculated, over the period of record. For each composite-station record, the interquartile range of monthly differences was estimated. To identify unusual values, the following relationship was evaluated: $dT_i - q50 > F (q75 - q25)$ where dT_i is an absolute difference between an estimated LW climatology value and a composite-station value for month i ; $q50$, $q75$, and $q25$ are the median, 75th and 25th percentiles of the differences for a station, over the period of record. F is a multiplier and it is set to 4.6 for air temperature which reduces the number of outliers (cf, Eischeid et al., 1995). If the difference between dT_i and $q50$ was greater than 4.6 times the interquartile range, the monthly value was considered to be an outlier. Values identified as outliers in this way comprised about 0.08% of the population of observations and were excluded from the composite-station record. With outliers removed, the overall number of available monthly station T values ranges from about 3,000 to 19,000.

SPATIAL INTERPOLATION:

Monthly averages of station air temperature (T) were interpolated to a 0.5-degree by 0.5-degree latitude/longitude grid, where the grid nodes were centered on the 0.25 degree. The gridded fields were estimated from monthly station averages using a combination of spatial interpolation methods: digital-elevation-model (DEM) assisted interpolation (Willmott and Matsuura, 1995); traditional interpolation (Willmott et al., 1985); and climatologically aided interpolation (CAI) (Willmott and Robeson, 1995). A climatology of T for each month at each station was produced first by combining average monthly station values from two available climatologies (described below). DEM-assisted interpolation then was used to estimate average monthly T s (climatology) at unsampled locations, principally at the grid nodes. Individual monthly gridded T fields were estimated next by using CAI (described below).

For the background climatology, two station climatologies were merged. The first was calculated at those of our air-temperature time-series stations which had at least ten years of observations for each month. The second was the monthly station T climatology of Legates and Willmott (1990). The stations in these two climatologies were merged to form the background climatology for CAI when the stations are located within 2.5 km of each other.

Traditional interpolation was accomplished with a spherical version of Shepard's algorithm, which employs an enhanced distance-weighting method (Shepard, 1968; Willmott et al., 1985). Our traditional interpolations of estimated sea-level T s, within our DEM-assisted procedure, as well as our interpolations of deviations from climatology, within CAI, were made in this way. The number of nearby stations that influence a grid-node estimate, however, was increased to an average of 20, from an average of 7 in earlier applications. This resulted in smaller cross-validation errors (see below) and visually more realistic air-temperature fields. A more robust neighbor finding algorithm, based on spherical distance, also was used.

Incorporating elevational influences, through an average air-temperature lapse rate, can increase the accuracy of spatially interpolating average air temperature (Willmott and Matsuura, 1995). DEM-assisted interpolation of average-monthly air temperature, therefore, was employed. Briefly, each average-monthly station air temperature was first brought down to sea level (warmed) at an average environmental lapse rate (6.0 °C/km). Traditional interpolation to the grid nodes then was performed on the adjusted-to-sea-level average-monthly station air temperatures. Finally, the gridded sea-level air temperatures were brought up to the DEM-grid height (cooled); once again, at the average environmental lapse rate.

Using a relatively high-resolution climatology also can increase the accuracy of spatially interpolated time series of monthly climate variables. Employing CAI (Willmott and Robeson, 1995), a monthly T at each time-series station was differenced from a climatologically averaged T for that month which was available at or was interpolated to the time-series station location. Traditional interpolation then was performed on the station differences to obtain a gridded difference field. Finally, the gridded difference field was added to the interpolated (DEM-assisted) estimates of the climatology at the same set of grid points.

SPATIAL CROSS VALIDATION:

To indicate (roughly) the spatial interpolation errors, station-by-station cross validation was employed (Willmott and Matsuura, 1995). One station was removed at a time, and the air temperature was then interpolated to the removed station location from the surrounding nearby stations. The difference between the real station value and the interpolated value is a local estimate of interpolation error. After each station cross validation was made, the removed station was put back into the network. To reduce network biases on cross-validation results, absolute values of the errors at the stations were interpolated to the same spatial resolution as the air temperature field.

ARCHIVE STRUCTURE:

air_temp_2017.tar.gz: Monthly-mean air temperatures for the years 1900-2017 interpolated to a 0.5 by 0.5 degree grid resolution (centered on 0.25 degree). The format of each record is:

Field	Columns	Variable	Fortran Format
1	1 - 8	Longitude (decimal degrees)	F8.3
2	9 - 16	Latitude (decimal degrees)	F8.3
3-14	17 - 112	Monthly Air Temperature (°C, Jan-Dec)	12F8.1

air_temp_cv2017.tar.gz: Cross-validation errors (absolute values) associated with air temperatures for the years 1900-2017 interpolated to a 0.5 by 0.5 degree grid resolution. The format of each record is:

Field	Columns	Variable	Fortran Format
1	1 - 8	Longitude (decimal degrees)	F8.3
2	9 - 16	Latitude (decimal degrees)	F8.3
3-14	17 - 112	Cross-validation errors (absolute values) of Monthly Temperature (°C, Jan-Dec)	12F8.1

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